Day 6 assignment

21 May 2026 20:49

Target service/process: docker

Environment Basics

Check Kernel & OS details:

uname -a

```
root@030b320cc2b0:/# uname -a
Linux 030b320cc2b0 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x8
6_64 GNU/Linux
```

Confirms architechture & kernal version before troubleshooting deeper issues

Check distribution Information:

cat /etc/os-release

```
root@030b320cc2b0:/# cat /etc/os-release
VERSION="26.04 (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
```

Useful for checking compatibility issues or package-related bugs

Filesystem Sanity Checks

Create temporary working directory:

mkdir -p /tmp/runbook-demo

```
root@030b320cc2b0:~# mkdir /tmp/runbook-demo
root@030b320cc2b0:~# ls
root@030b320cc2b0:~# cd tmp
bash: cd: tmp: No such file or directory
root@030b320cc2b0:~# cd /
root@030b320cc2b0:/# cd tmp
root@030b320cc2b0:/tmp# ls
runbook-demo
```

Temporary troubleshooting workspace created, also confirms filesystem is writable

Copy & Verify File Operations

cp /etc/hosts /tmp/runbook-demo/hosts-copy && ls -l /tmp/runbook-demo

```
root@030b320cc2b0:/# cp /etc/hosts /tmp/runbook-demo/hosts-copy && ls -l tmp/runbook-demo
total 4
-rw-r--r-- 1 root root 172 May 21 15:48 hosts-copy
```

File copy succeeded - permissions & storage access appear normal

Snapshot Docker Process Resource Usage

ps -o pid,pcpu,pmem,comm -C dockerd

```
jeenicj@DESKTOP-BG3MAVI:/$ ps -o pid,pcpu,pmem,comm -C dockerd
    PID %CPU %MEM COMMAND
    387  0.0  1.3 dockerd
```

Docker daemon is active with low usage - no immediate resource exhaustion detected

Check System Memory:

free -h

```
root@030b320cc2b0:/# free -h
               total        used        free      shared  buff/cache   available
Mem:           3.6Gi       1.0Gi       306Mi       7.1Mi       2.3Gi       2.6Gi
Swap:          1.0Gi        64Mi       959Mi
```

Healthy available memory - swap usage is minimal, indicating no memory pressure currently

Disk & IO

Check Disk Usage:

df -h

```
root@030b320cc2b0:/# df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay        1007G  7.5G  949G   1% /
tmpfs            64M     0   64M   0% /dev
shm              64M     0   64M   0% /dev/shm
/dev/sde       1007G  7.5G  949G   1% /etc/hosts
tmpfs           1.9G     0  1.9G   0% /proc/acpi
tmpfs           1.9G     0  1.9G   0% /proc/scsi
tmpfs           1.9G     0  1.9G   0% /sys/firmware
```

Root filesystem has sufficient free space - no immediate disk exhaustion risk

du -sh /var/log

```
root@030b320cc2b0:/# du -sh /var/log
416K    /var/log
```

Logs consume moderate storage - worth monitoring if logs growing rapidly

Network

Check Listening Ports

ss -xlpn | grep docker (x-> unix sockets) OR ss -tulpn (t,u -> tcp/udp)

```
jeenicj@DESKTOP-BG3MAVI:/mnt/c/Users/Jeeni$ ss -xlpn | grep docker
u_str LISTEN 0      4096                                   /var/run/docker/metrics.sock 4766
        * 0
u_str LISTEN 0      4096                           /var/run/docker/libnetwork/156179e98aef.sock 9550
        * 0
u_str LISTEN 0      4096   /mnt/wsl/docker-desktop/shared-sockets/guest-services/distro-services/ubuntu.sock 14019
        * 0
u_str LISTEN 0      4096                                               /run/docker.sock 7835
        * 0
u_str LISTEN 0      4096                                           /var/run/docker.sock 10225
        * 0
u_str LISTEN 0      4096                                       /var/run/docker-cli.sock 14018
        * 0
```

Docker daemon is listening on expected port/interface - confirms service networking is active

Verify Service Connectivity

curl --unix-socket /var/run/docker.sock http://localhost/_ping

```
jeenicj@DESKTOP-BG3MAVI:/mnt/c/Users/Jeeni$ curl --unix-socket /var/run/docker.sock http://localhost/_ping
OKjeenicj@DESKTOP-BG3MAVI:/mnt/c/Users/Jeeni$
```

(curl -I http://localhost:2375/_ping -> **Docker does not expose its API over TCP port 2375**. Instead, it listens on a **Unix socket** (/var/run/docker.sock))

Docker API endpoint responds successfully - service reachable locally without timeouts/errors

Logs Reviewed

Review Docker Service Logs

journalctl -u docker -n 50

```
jeenicj@DESKTOP-BG3MAVI:/mnt/c/Users/Jeeni$ journalctl -u docker -n 50
May 20 13:01:41 DESKTOP-BG3MAVI dockerd[451]: time="2026-05-20T13:01:41.078370955Z" level=info msg="Daemon shutdown com>
May 20 13:01:41 DESKTOP-BG3MAVI systemd[1]: docker.service: Deactivated successfully.
May 20 13:01:41 DESKTOP-BG3MAVI systemd[1]: Stopped docker.service - Docker Application Container Engine.
May 20 13:01:41 DESKTOP-BG3MAVI systemd[1]: docker.service: Consumed 3.988s CPU time.
-- Boot de034a7a6a9f4c359a91bfa5a827890e --
May 20 13:08:28 DESKTOP-BG3MAVI systemd[1]: Starting docker.service - Docker Application Container Engine...
May 20 13:08:29 DESKTOP-BG3MAVI dockerd[364]: time="2026-05-20T13:08:29.011778369Z" level=info msg="Starting up"
May 20 13:08:29 DESKTOP-BG3MAVI dockerd[364]: time="2026-05-20T13:08:29.015852218Z" level=info msg="OTEL tracing is not>
May 20 13:08:29 DESKTOP-BG3MAVI dockerd[364]: time="2026-05-20T13:08:29.016837851Z" level=info msg="CDI directory does >
May 20 13:08:29 DESKTOP-BG3MAVI dockerd[364]: time="2026-05-20T13:08:29.016895239Z" level=info msg="CDI directory does >
May 20 13:08:29 DESKTOP-BG3MAVI dockerd[364]: time="2026-05-20T13:08:29.103500593Z" level=info msg="Creating a containe>
May 20 13:08:29 DESKTOP-BG3MAVI dockerd[364]: time="2026-05-20T13:08:29.167618657Z" level=info msg="Loading containers:>
May 20 13:08:29 DESKTOP-BG3MAVI dockerd[364]: time="2026-05-20T13:08:29.284679143Z" level=info msg="[graphdriver] using>
May 20 13:08:29 DESKTOP-BG3MAVI dockerd[364]: time="2026-05-20T13:08:29.332771026Z" level=info msg="Restoring container>
May 20 13:08:29 DESKTOP-BG3MAVI dockerd[364]: time="2026-05-20T13:08:29.543004683Z" level=info msg="Deleting nftables I>
```

No critical errors in recent logs - service startup appears healthy

Review System Log Tail

tail -n 50 /var/log/syslog

```
jeenicj@DESKTOP-BG3MAVI:/mnt/c/Users/Jeeni$ tail -n 50 /var/log/syslog
2026-05-22T02:08:14.135024+00:00 DESKTOP-BG3MAVI microk8s.daemon-kubelite[568964]: I0522 02:08:14.134977  568964 secure_
serving.go:213] Serving securely on [::]:16443
2026-05-22T02:08:14.135210+00:00 DESKTOP-BG3MAVI microk8s.daemon-kubelite[568964]: I0522 02:08:14.135064  568964 tlsconf
ig.go:243] "Starting DynamicServingCertificateController"
2026-05-22T02:08:14.135258+00:00 DESKTOP-BG3MAVI microk8s.daemon-kubelite[568964]: I0522 02:08:14.135128  568964 local_a
vailable_controller.go:156] Starting LocalAvailability controller
2026-05-22T02:08:14.135281+00:00 DESKTOP-BG3MAVI microk8s.daemon-kubelite[568964]: I0522 02:08:14.135143  568964 cache.g
o:32] Waiting for caches to sync for LocalAvailability controller
2026-05-22T02:08:14.135317+00:00 DESKTOP-BG3MAVI microk8s.daemon-kubelite[568964]: I0522 02:08:14.135194  568964 dynamic
_serving_content.go:135] "Starting controller" name="aggregator-proxy-cert::/var/snap/microk8s/8702/certs/front-proxy-cl
ient.crt::/var/snap/microk8s/8702/certs/front-proxy-client.key"
```

No kernel panic, OOM, or disk-related warnings observed - general system logs appear stable

Quick Findings

- Docker service is running normally.
- CPU and memory usage are low.
- Disk space is healthy with no filesystem concerns.
- Network connectivity to the Docker API is successful.
- No major errors detected in logs.
- System appears stable at the time of investigation.

If this worsens

1. Restart Strategy

Sudo systemctl restart docker
Sudo systemctl start docker

Restart docker service if containers stop responding or API becomes unavailable

2. Increase Logging & Monitor Live Events

Journalctl -u docker -f
Docker events

Monitor real-time logs and container lifecycle events for intermittent failures

3. **Strace -p <dockerd-pid>**
   Top
   Iotp

Use tracing & performace tools if CPU spikes, hands, or IO bottlenecks occur